

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:100

Annexure A

Industry Problem Solving

Code of Conduct:

*I **Chalamala Sai Harshitha (192424346)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

C. Sai Harshitha

Annexure A

1. A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.

1. Proposed Solution

The system can use customer purchase information such as:

- Products previously purchased
- Purchase frequency, Quantity purchased
- Product categories, Spending patterns

This information is represented as features for each customer. When a customer requires a recommendation, KNN identifies customers with similar purchasing patterns.

2. Measuring Similarity

Similarity between customers can be measured using **Euclidean distance**.

Euclidean Distance:

$$d = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2]}$$

where:

- x = features of the target customer
- y = features of another customer
- d = distance between the two customers

3. Generating Recommendations

The system selects the **K nearest customers** to the target customer.

For example, if $K = 3$:

1. Calculate the distance between the target customer and other customers.
2. Select the three customers with the smallest distances.
3. Identify products purchased by these similar customers.
4. Remove products that the target customer has already purchased.
5. Recommend products that are frequently purchased among the nearest customers.

Example:

Customer	Purchased Products
A	Laptop, Mouse
B	Laptop, Mouse, Keyboard
C	Laptop, Keyboard, Headset
D	Laptop, Mouse, Keyboard

If Customer A is the target customer, Customers B, C, and D may be identified as similar

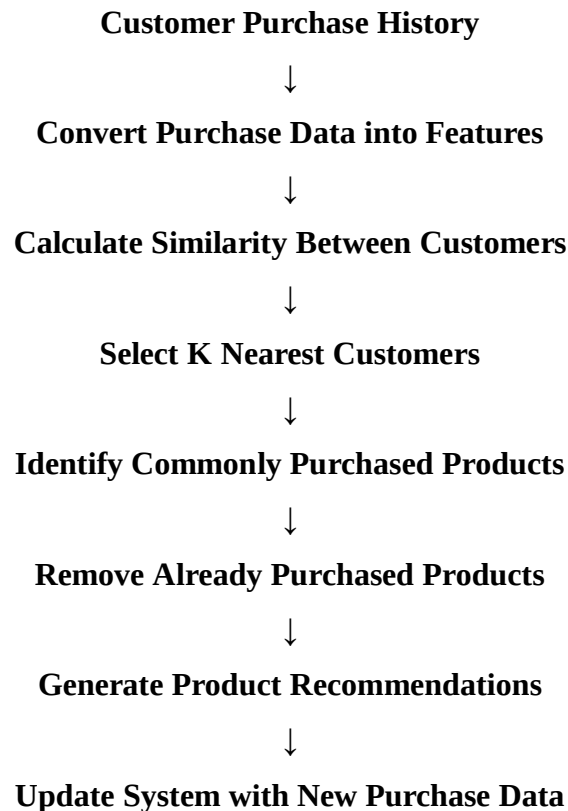
customers. Since **Keyboard** is purchased by several similar customers, it can be recommended to Customer A.

4. Adapting to New Customer Data

The system can adapt whenever new purchase information becomes available. When a customer makes a new purchase, the updated purchase information is added to the system.

The customer's feature representation is then updated, allowing KNN to find more relevant neighbours for future recommendations.

6. Overall Working



Conclusion

A **KNN-based recommendation system** is suitable for the retail company's problem because it identifies customers with similar purchasing behaviour and uses their purchase history to generate personalized recommendations. Similarity can be measured using Euclidean distance, recommendations can be generated from the nearest customers, and the system can adapt as new customer purchase data becomes available.

2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.

1. Proposed Solution

The system collects sensor readings such as:

- Temperature, Pressure
- Vibration, Speed
- Operating time

The **machine output** is considered the target value. When a new sensor reading is received, LWR uses previous observations that are most similar to the current condition to predict the output.

2. Working of Locally Weighted Regression

For a new sensor reading, LWR assigns a weight to each existing data point.

- Similar or nearby data points → **Higher weight**
- Dissimilar or distant data points → **Lower weight**

A commonly used weight formula is:

$$w_i = e^{[-(x-x_i)^2 / (2\tau^2)]}$$

where:

- w_i = weight assigned to the data point
- x = current sensor reading
- x_i = previous sensor reading
- τ = bandwidth parameter

The weighted observations are then used to build a local regression model and predict the machine output.

3. How Local Models Improve Prediction Accuracy

A single global regression model may not accurately represent machine behaviour under different operating conditions.

LWR creates a **local model around the current sensor conditions**. Therefore:

- Relevant observations have greater influence.
- Changes in machine behaviour can be captured.
- Different operating conditions can have different local relationships.
- Predictions can be more accurate when machine behaviour changes over time.

For example, if machine output changes differently at low and high temperatures, LWR focuses more on historical observations with temperatures similar to the current temperature.

4. Handling Noisy Data

Sensor readings may contain noise because of measurement errors or temporary fluctuations.

The system can handle noisy data by:

- Giving lower weights to less relevant observations.
- Detecting and reducing the effect of extreme outliers.
- Applying suitable preprocessing or smoothing techniques.

Thus, irrelevant noisy observations have less influence on the local prediction.

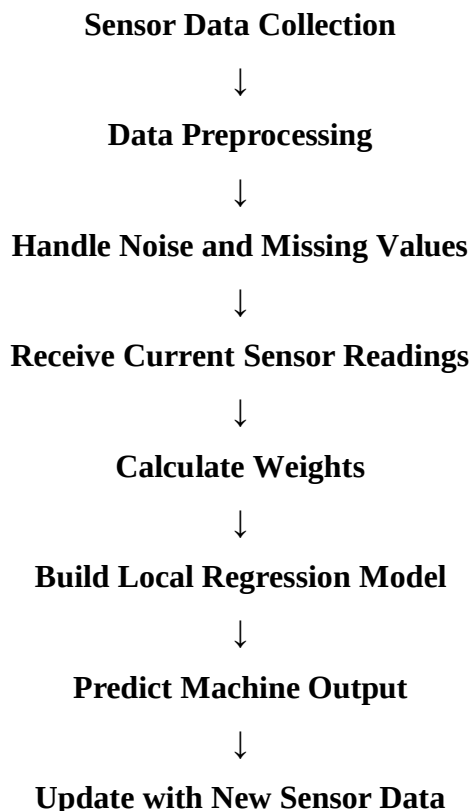
5. Handling Incomplete Data

Some sensor readings may be missing. Before applying LWR, missing values can be handled using appropriate preprocessing techniques such as:

- Mean or median replacement
- Interpolation for time-based readings
- Using available features when appropriate

This helps the prediction system continue working even when some sensor values are unavailable.

6. Overall Workflow



Conclusion

Locally Weighted Regression is suitable for this manufacturing problem because it focuses on observations that are most relevant to the current sensor conditions.

3. A healthcare company aims to classify patients into risk categories using historical medical data. Suggest a model using KNN or RBF networks. Explain how the algorithm handles large datasets, feature selection, and accuracy improvement in real-world scenarios.

1. Proposed Solution

The system can use historical patient data containing features such as:

- Age, BMI
- Blood pressure, Blood sugar level
- Cholesterol level

The target output can be a risk category such as **Low Risk, Medium Risk, or High Risk**.

For a new patient, KNN compares the patient's features with those of previously recorded patients.

2. Working of KNN

The system calculates the distance between the new patient and existing patients.

Euclidean Distance:

$$d = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2]}$$

A smaller distance indicates greater similarity.

The system then selects the **K nearest patients** and checks their risk categories. The new patient is assigned the category that receives the majority of votes.

For example, if $K = 5$ and three of the five nearest patients are classified as **High Risk**, the new patient can be classified as High Risk.

3. Handling Large Datasets

KNN can become computationally expensive when the healthcare dataset is very large because distances need to be calculated against many stored records.

The system can improve efficiency by:

- Using suitable data structures for faster nearest-neighbour searching.
- Removing unnecessary features and records.
- Using an appropriate value of K .
- Applying dimensionality reduction when the number of features is very high.

4. Feature Selection

Not every medical feature contributes equally to risk classification. Irrelevant or redundant features can reduce model performance.

Feature selection can identify the most useful medical attributes, such as:

- Blood pressure, Blood sugar, Cholesterol, Age, BMI

Using relevant features can reduce computation and help KNN make more meaningful comparisons between patients.

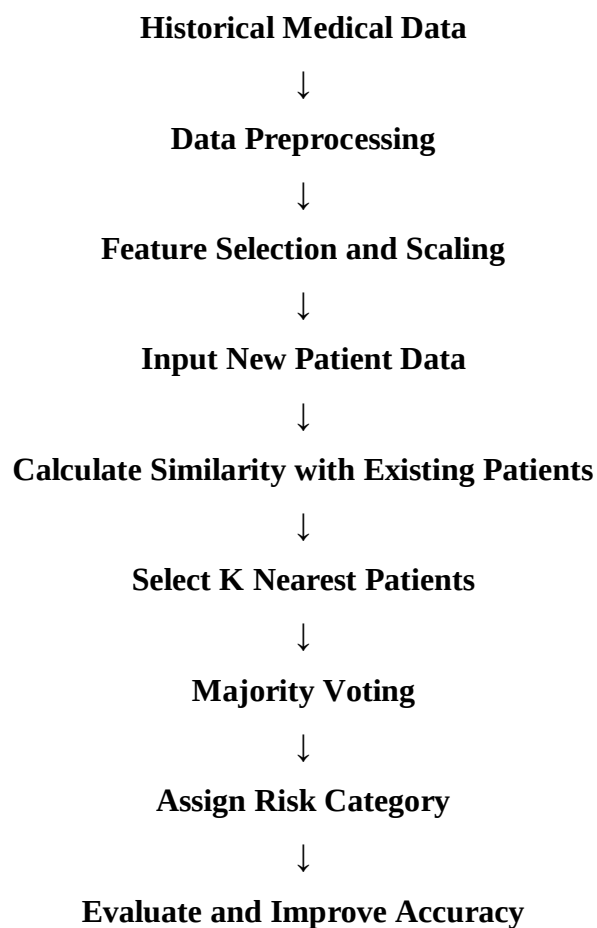
5. Improving Accuracy

The accuracy of the system can be improved by:

- Choosing an appropriate value of K.
- Normalizing or standardizing numerical features.
- Selecting relevant medical features.
- Handling missing values appropriately.
- Removing or reducing the effect of abnormal noisy data.
- Evaluating the model using suitable test data.

Feature scaling is particularly important because medical attributes may have very different numerical ranges.

6. Overall Working



Conclusion

A KNN-based healthcare risk classification system can classify new patients by comparing their medical features with similar patients from historical data.

4. An e-commerce platform wants to detect similar products for dynamic pricing strategies. Design a solution using Case-Based Learning. Discuss how similarity metrics are defined, how past cases are reused, and how the system scales with increasing product data.

1. Proposed Solution

Each product can be represented using features such as:

- Product category, Brand
- Price, Product rating, Sales volume
- Customer demand, Product popularity

The system stores these details along with previous pricing decisions and their outcomes.

When a new product is added, the system searches for **similar products from previous cases**.

2. Defining Similarity Metrics

Similarity can be calculated by comparing the important product features.

For numerical features, **Euclidean distance** can be used:

$$d = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2]}$$

A smaller distance indicates that two products are more similar.

For categorical features such as product category or brand, matching-based similarity can be used.

The system can assign different importance to different features. For example, product category and customer demand may be given higher importance because they strongly influence pricing.

3. Reusing Past Cases

After finding similar products, the system examines their previous pricing cases.

For example:

Product	Demand	Rating	Previous Price	Result
A	High	4.5	₹1,000	High Sales
B	High	4.4	₹950	High Sales
C	Low	3.8	₹800	Low Sales

If a new product has characteristics similar to Products A and B, the system can reuse their pricing information when determining a suitable price.

The previous cases therefore provide practical information for making new pricing decisions.

4. Dynamic Pricing Process

The system can follow these steps:

1. Collect information about the new product.
2. Compare it with stored product cases.
3. Calculate similarity between the new product and previous products.
4. Retrieve the most similar cases.

5. Examine their previous prices and sales outcomes.
6. Select or adjust a suitable price based on those cases.
7. Store the new product and its pricing outcome as a new case.

This allows the pricing strategy to change according to product characteristics and market conditions.

5. Scaling with Increasing Product Data

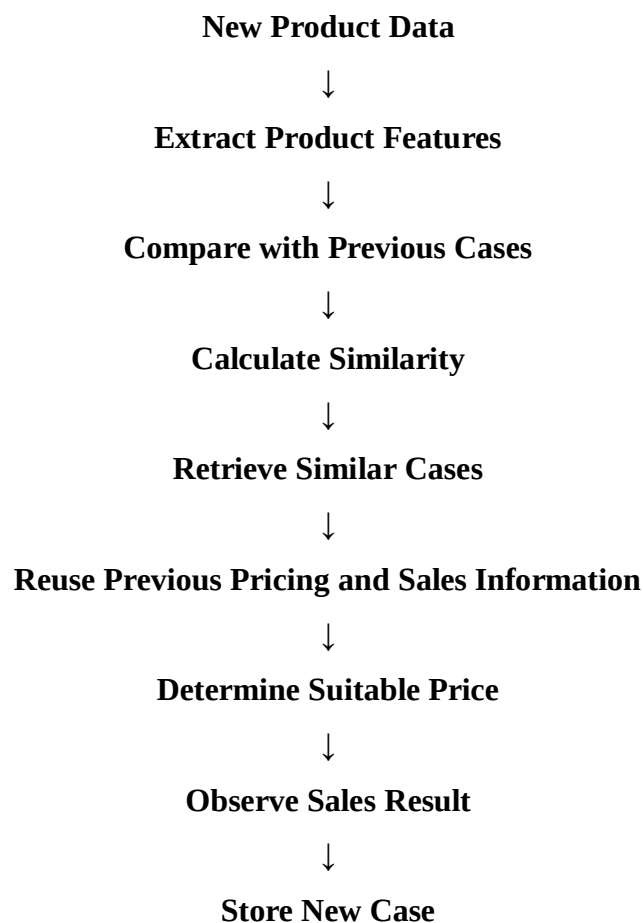
As the number of products increases, searching through every stored case can become slower.

The system can improve scalability by:

- Selecting only important product features.
- Storing cases in an organized manner.
- Retrieving only the most relevant cases.
- Removing outdated or duplicate cases when appropriate.
- Using efficient similarity-search techniques for large datasets.

These approaches reduce unnecessary comparisons and help the system handle increasing product data.

6. Overall Working



Conclusion

Case-Based Learning is suitable for the e-commerce platform because it allows the system to identify products similar to a new product and reuse knowledge from previous pricing cases.

5. A logistics company needs to forecast delivery times based on location and traffic patterns. Propose a solution using Radial Basis Function networks or Locally Weighted Regression. Explain how the model captures non-linear relationships and improves prediction reliability under varying conditions.

1. Proposed Solution

The system can use input features such as:

- Delivery location, Distance
- Traffic level, Time of day
- Day of the week

The **delivery time** is used as the target output.

The RBF network learns the relationship between these input factors and historical delivery times.

2. Working of RBF Network

An RBF network generally consists of three layers:

1. **Input Layer** – receives location, distance, traffic, and other relevant information.
2. **Hidden Layer** – uses radial basis functions to measure how close the input is to learned patterns.
3. **Output Layer** – predicts the expected delivery time.

The radial basis function can be represented as:

$$\phi(x) = e^{-||x-c||^2 / (2\sigma^2)}$$

where:

- x = current input
- c = centre of the radial basis function
- σ = spread of the function
- $\phi(x)$ = activation value

Inputs that are closer to a learned centre receive a higher activation.

3. Capturing Non-Linear Relationships

Delivery time does not always increase in a simple linear manner.

For example, a small increase in traffic may have little effect during normal conditions but may cause a large increase in delivery time during peak hours.

The RBF network can learn such complex relationships between:

Location + Distance + Traffic + Time Conditions → Delivery Time

Therefore, it can provide better predictions when delivery patterns are non-linear.

4. Handling Varying Conditions

Traffic and road conditions can change throughout the day. The system can improve prediction reliability by continuously using updated delivery information.

For example:

- **Low traffic** → shorter predicted delivery time
- **Moderate traffic** → moderate predicted delivery time
- **Heavy traffic** → longer predicted delivery time

Historical delivery data under different conditions helps the RBF network learn these patterns.

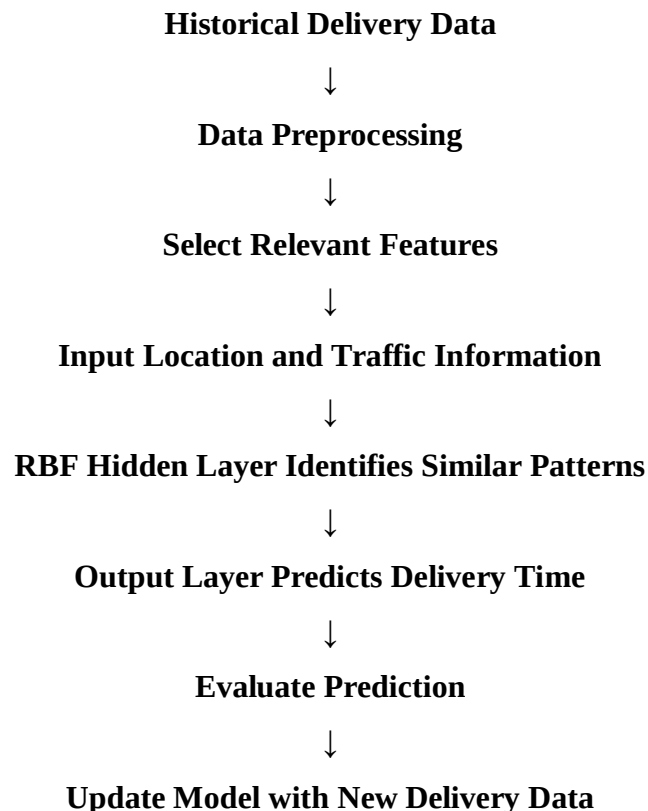
5. Improving Prediction Reliability

The system can improve reliability by:

- Using sufficient historical delivery data.
- Selecting relevant input features.
- Normalizing input values.
- Handling missing and noisy data.
- Updating the model with recent delivery information.
- Testing the model on unseen delivery data.

These steps help the model produce more reliable predictions under changing traffic and location conditions.

6. Overall Working



Conclusion

An **RBF network** provides a suitable solution for forecasting delivery times because it can learn complex, non-linear relationships between location, traffic patterns, and delivery time.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand